

Week 2

Section 1.1-1.6

Work practice problems

- mechanics, stats

- newtons laws

- two new units of measurement

- Develop procedures for numerical calculations

SI $\rightarrow$ international system

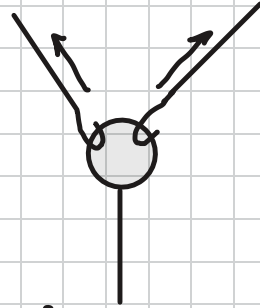
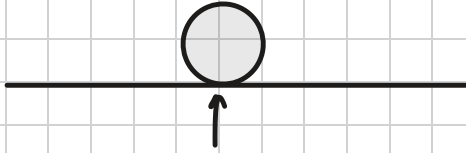
SI

What is mechanics?

What happens to a "body" when a force is applied

extremely small deformation

rigid body



We study any body at rest (Static) or in motion (Dynamic)

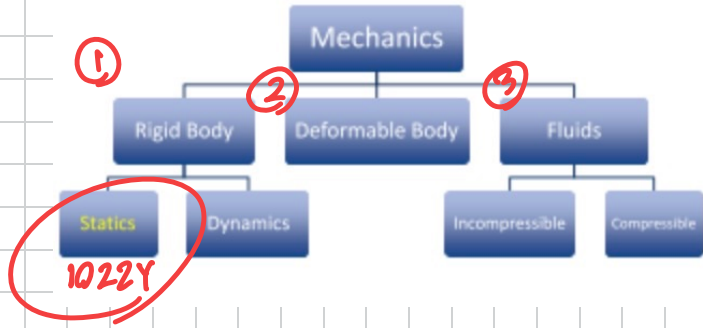
- rigid body
 - deformable body
 - fluid
- } different mechanics

Statics vs dynamics

deals with
body at
rest or in
constant
motion

deals with
body in accel
eration

we will be dealing
with static rest



types of forces encountered

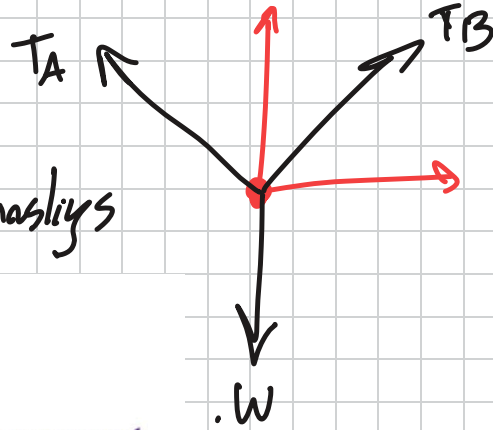
- Self weight static
- vehicles
- people static
- wind dynamic
- waves dynamic
- seismic activity dynamic

Basic Quantities

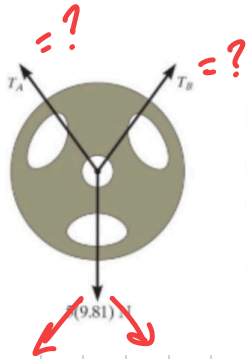
- ① Length point in space
- ② Time statistics is time independent
- ③ Mass not Force
- ④ Force vector quantity, Push or pull

Particle: body of negligible dimension

- can be analyzed as a point mass
- geometry is not considered in analysis



$$W = m \cdot g$$

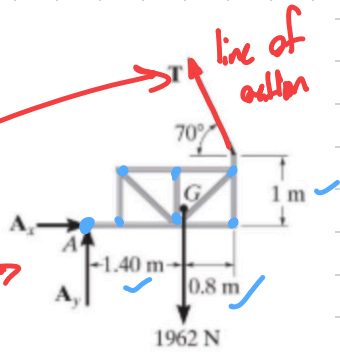
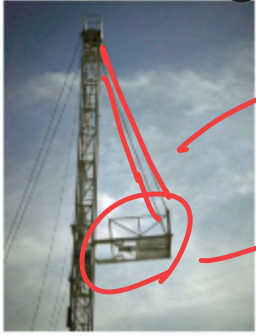


$$5 \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

Forces are concurrent
(the lines of action of all forces intersect at a common point)

rigid body

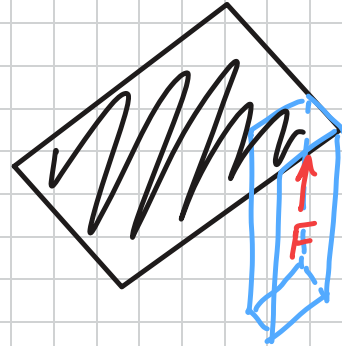
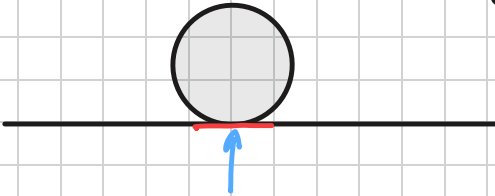


Forces are not
Concurrent

Forces either
external or internal

Cannot idealize as a
particle

Concentrated force
acts on a point on
the body



Newton's Laws of motion ★

- ① a particle will remain at rest or continue to move in a straight line at constant velocity
- ② a body with unbalanced force will experience acceleration in the direction of the force
$$\vec{F} = m\vec{a}$$
- ③ for every action there is an equal and opposite reaction



Pen not working !!

$$\left(\frac{\cancel{4716}}{\cancel{\text{in}}} \right) \left(\frac{4448 \text{ N}}{\cancel{1 \text{ Lb}}} \right) \left(\frac{\cancel{1 \text{ ft}}}{\cancel{12 \text{ in}}} \right) \left(\frac{0.3048 \text{ m}}{\cancel{1 \text{ ft}}} \right) = \text{N/m}$$

wait I think I fixed it..

Slug unit

- mass
- FPS unit
- $\left(\frac{\text{Lb} \cdot \text{s}^2}{\text{ft}} \right)$

Law of gravity

$$F = G \frac{m_1 m_2}{r^2}$$

Gravitational Attraction

weight
↳ $W = G \frac{mM_e}{r^2}$ Object near earth surface

$$g = GM_e / r^2$$

radius from earth center to particle

If using
slug

↳

$$g = 32.2 \text{ ft/s}^2$$

in US units

$$\longrightarrow 386.4 \text{ in/s}^2$$

$$D = V \times T$$

Diagram showing the relationship between variables and their units:

- D points to (m)
- V points to $m/3$
- T points to s

$$Lhs = Rh3$$

3 Sig figs